MAT 5173 — Weekly Schedule

Algebra I · Fall 2026 · UT San Antonio

Texts

Primary text

- **Ash** — Ash, *Abstract Algebra: The Basic Graduate Year*

Secondary reading

- **LiZh** — Li & Zhao, *Introduction to Abstract Algebra*
- **CLO** — Cox, Little & O'Shea, *Ideals, Varieties, and Algorithms*

Supplementary reading

- **Shafarevich** — Shafarevich, *Basic Algebraic Geometry 1*
- **Rosebrock** — Rosebrock, *Visual Group Theory*
- **Beardon** — Beardon, *Algebra and Geometry*

Handouts & papers

- **Smith** — Smith, Kahanpaa, Kekalainen & Traves, *An Invitation to Algebraic Geometry*

Schedule

Date	Topic	Reading
Wed 19 Aug	Lecture — Course launch; groups and subgroups	Ash §1.1; LiZh §1.3-1.4
Mon 24 Aug	Lecture — Permutation groups; cyclic groups; the order of an element	Ash §1.2; LiZh §1.5, §2.1; Rosebrock Ch. 2 (source of the symmetry-group examples)
Wed 26 Aug	Lecture — Cosets, normal subgroups and homomorphisms	Ash §1.3; LiZh §3.1, §3.3
Mon 31 Aug	Lecture — The isomorphism theorems	Ash §1.4; LiZh §3.4
Wed 2 Sep	Lecture — Direct products	Ash §1.5; Beardon Ch. 12 (worked examples in symmetry groups)
Wed 9 Sep	**Q&A** (**Due**: Set 1)	
Mon 14 Sep	Lecture — Rings: basic definitions and properties	Ash §2.1; LiZh §4.1
Wed 16 Sep	Lecture — Ideals, homomorphisms and quotient rings; the isomorphism theorems for rings	Ash §2.2-2.3; LiZh §4.5
Mon 21 Sep	Lecture — Maximal and prime ideals	Ash §2.4; LiZh §4.7
Wed 23 Sep	Lecture — Polynomial rings; unique factorization	Ash §2.5-2.6; LiZh §5.1

Date	Topic	Reading
Mon 28 Sep	Lecture — Principal ideal domains and Euclidean domains; rings of fractions; irreducible polynomials	Ash §2.7-2.9; LiZh §5.2-5.3
Wed 30 Sep	**Q&A** (**Due**: Set 2)	
Mon 5 Oct	**Exam** — Midterm Exam 1 — Sets 1 and 2	
Wed 7 Oct	Lecture — Field extensions; degree; simple extensions	Ash §3.1; LiZh §6.1-6.2
Wed 14 Oct	Lecture — Splitting fields; algebraic closures	Ash §3.2-3.3; LiZh §6.3
Mon 19 Oct	Lecture — Separability; normal extensions; finite fields	Ash §3.4-3.5; Ash §6.4; LiZh §6.4
Wed 21 Oct	Lecture — Affine space and affine varieties	CLO §1.1-1.2; Ash §8.1; Smith Ch. 1 (handout)
Mon 26 Oct	Lecture — V(I) and I(V): the two maps	CLO §1.4; Ash §8.1
Wed 28 Oct	Lecture — The coordinate ring; the algebra-geometry dictionary; the Nullstellensatz stated, and why k must be algebraically closed	CLO §1.4; Ash §8.3
Mon 2 Nov	**Q&A** (**Due**: Set 3)	
Wed 4 Nov	Lecture — The Hilbert basis theorem; Noetherian rings; every variety is cut out by finitely many equations	Ash §8.2
Mon 9 Nov	Lecture — Plane curves: rational curves, the node and the cusp	Shafarevich §1.1
Wed 11 Nov	Lecture — Fixed fields and Galois groups	Ash §6.1
Mon 16 Nov	Lecture — The fundamental theorem of Galois theory	Ash §6.2; Ash §6.3
Wed 18 Nov	Lecture — Solvability by radicals: the unsolvable quintic — a survey	Ash §6.8
Mon 23 Nov	**Q&A** (**Due**: Set 4)	
Mon 30 Nov	*Review* — Review — the exam pool	
Wed 2 Dec	**Exam** — Midterm Exam 2 — Sets 3 and 4	